

Revised Generation Interconnection System Impact Study Report

For Queue Project Q20-1568

Websterville 100 kV
July 22, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1568. The Interconnection Customer proposes a 264.3 MW wind facility interconnecting at the Websterville 100 kV substation in the Indiana area. The request was received on September 09, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 605 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 12 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$39,337,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
WEST MOULTON / LEYENDECKER 100 kV L	LEYENDECKER / ST MARYS 100 kV	161.9	129	9.1
WEST MOULTON / GLENCLIFF 100 kV	LEYENDECKER / ST MARYS 100 kV	161.0	122	9.1
WEBSTERVILLE / POWERLINE ELKHORN 100 kV Line	THOUSAND PRINGS / BLOUNT 100 kV	147.2	199	20.0
LEYENDECKER / ST MARYS 100 kV	THOUSAND SPRINGS / HANSEL 100 kV	142.0	147	13.5
GLENCLIFF / LENKURT 100 kV line	ST MARYS / LEYENDECKER 100 kV	126.1	112	5.3
SPEARMAN CITY / EDES TAP 1 100 kV	DUTTON PUMP / LEYENDECKER 100 kV L	122.5	119	7.8
DUTTON PUMP / LEYENDECKER 100 kV	MUSKOGEE / MONTPELIER ELECTRIC GENERATING STA 500 kV	120.8	256	12.5
EDES TAP 1 / HATO REY GIS 1403-1404 100 kV	DUTTON PUMP / LEYENDECKER 100 kV	119.6	118	7.8
SNYDER / EASTMONT 100 kV	THOUSAND SPRINGS / HANSEL 100 kV	116.1	125	30.4
DON / HATO REY GIS 1403-1404 100 kV	DUTTON PUMP SUBSTATION / LEYENDECKER 100 kV	113.6	111	7.8
HENRY COUNTY 500-230 kV Transformer Bank	LONETREE / EAST METHUEN 500 kV L	104.6	1,383	19.8
TREGO / WEST MELROSE 230 kV line	SWAN VLY SUBSTATION / TREGO 500 kV	103.4	623	5.7

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
WEST MOULTON / LEYENDECKER 100 kV Line	Rebuild line with higher-capacity conductor	\$5,974,000	59% (\$3,505,000)
WEST MOULTON / GLENCLIFF 100 kV	Rebuild line with higher-capacity conductor	\$4,847,000	59% (\$2,845,000)
POWERLINE ELKHORN / WEBSTERVILLE 100 kV	Rebuild line with higher-capacity conductor	\$9,862,000	69% (\$6,765,000)
ST MARS SUBSTATION / LEYENDECKER 100 kV Line	Rebuild line with higher-capacity conductor	\$855,000	60% (\$514,000)
GLENCLIFF / LENKURT 100 kV line	Rebuild line with higher-capacity conductor	\$4,701,000	100% (\$4,701,000)
SPEARMAN CITY / EDES TAP 1 100 kV	Rebuild line with higher-capacity conductor	\$12,059,000	58% (\$7,056,000)
DUTTON PUMP / LEYENDECKER 100 kV	Rebuild line with higher-capacity conductor	\$9,964,000	58% (\$5,822,000)
EDES TAP 1 / HATO REY GIS 1403-1404 100 kV	Rebuild line with higher-capacity conductor	\$5,238,000	58% (\$3,065,000)
SNYDER / EASTMONT 100 kV	Rebuild line with higher-capacity conductor	\$3,431,000	63% (\$2,150,000)
DON / HATO REY GIS 1403-1404 100 kV	Rebuild line with higher-capacity conductor	\$1,990,000	58% (\$1,164,000)
HENRY COUNTY SUBSTATION 500/230kV Auto	Replace transformer with higher-rated unit	\$33,610,000	4% (\$1,522,000)
TREGO / WEST MELROSE 230 kV line	Reconductor line	\$6,155,000	4% (\$228,000)
	Total allocated to project		\$39,337,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.